Additional information

Approach

My overall goal was to use Pulp to create an optimization for the co-location model. It's a framework I've used before, so integration was easier. As any forecast produced, given a time frame, would have been extremely poor, I chose to go with a very simplistic historical average as the price forecast and focus on the optimization model.

I choose to not follow the specified price/value matrix because I think it makes more sense. The aFRR auction and solar production should be bid in with limitation prices because the aFRR has an alternative in the spot and the solar should only be sold if the spot is above the margin. For the DA arbitrage volumes however, using spot bids will decrease the risk of imbalances.

Assumptions

- The initial state of charge (SOC) on the battery is assumed to be 50%. there is no constraint on the final SOC.
- It is assumed that the activations given in capacity market will manifest, and therefore the cycle cost limit is also applied to capacity activations. Normally, you would look at the average energy activation and use that as a proxy for energy throughput.

Discussion points

- In reality, the AFR auction is so shallow that a 50 MW battery would have to consider how it affects the AFR price by bidding in. Therefore, you should look into splitting the capacity of the battery between multiple products or potentially putting in a bid ladder for your aFRR volumes.
- It is assumed that the battery will not have any charge or discharge as a result of the aFRR capacity clearing.
- Choose between Dayahead and Ancillary services, you need both the value of the capacity auction and the expected mean value of energy activations. Otherwise, the spread restrictions become very limiting.

Further improvements

- 25 EUR/MWh is high for a degradation cost and will really affect the operations of the battery. With multi-market operations having a position to start from is often more important and results in a higher P&L than sticking to a firm degradation cost.
- Probabilistic forecasting of the day-ahead price.

- When deciding whether an aFRR activation is attractive, you normally look at both the forecasted capacity price, and the forecasted activation volume and price to ensure you get a minimum spread.
- The model cannot do intraday trades, so ensuring that the aFRR activations can always be delivered is a big constraint on the model.
- The model of course is missing the all market optimization but the real interesting cases comes when looking into using the assets actively to mitigate imbalances.